Market Basket Analysis for Understanding Customer Purchase Behavior

A Proposal report for the BDM capstone Project

Submitted by

Sumit Kumar Jena

23f2002871

23f2002871@ds.study.iitm.ac.in



IITM Online BS Degree Program,

Indian Institute of Technology, Madras, Chennai

Tamil Nadu, India, 600036

Contents

1. Executive Summary and Title 4

2. Organisation Background 4

3. Problem Statement 5

    3.1 Problem statement 1 5

    3.2 Problem statement 2 5

4. Background of the Problem 5

5. Problem Solving Approach 6

    5.1 Methods 6

    5.2 Data Collection 6

    5.3 Analysis Tools 7

6. Expected Timeline 7

    6.1 Work Breakdown Structure 7

    6.2 Gantt Chart 8

7. Expected Outcome 8

Declaration Statement

I am working on a Project titled "**Market Basket Analysis for Understanding Customer Purchase Behavior**". I extend my appreciation to **Nana Grocery Store**, for providing the necessary resources that enabled me to conduct my project.

I hereby assert that the data presented and assessed in this project report is genuine and precise to the best of my knowledge and ability. The data has been gathered through primary sources and properly analyzed to assure its reliability.

Additionally, I affirm that all procedures employed for the purpose of data collection and analysis have been duly explained in this report. The outcomes and inferences derived from the data are an accurate depiction of the findings acquired through thorough analytical procedures.

I am dedicated to adhering to the information of academic honesty and integrity, and I am receptive to any additional examination or validation of the data contained in this project report.

I understand that the execution of this project is intended for individual completion and is not to be undertaken collectively. I thus affirm that I am not engaged in any form of collaboration with other individuals, and that all the work undertaken has been solely conducted by me. In the event that plagiarism is detected in the report at any stage of the project's completion, I am fully aware and prepared to accept disciplinary measures imposed by the relevant authority.

I agree that all the recommendations are business-specific and limited to this project exclusively, and cannot be utilized for any other purpose with an IIT Madras tag. I understand that IIT Madras does not endorse this.

Signature of Candidate:



Name: Sumit Kumar Jena
Date: 05/10/2025

1. Executive Summary and Title

Title: Market Basket Analysis for Understanding Customer Purchase Behavior

Nana Grocery Store is a small neighborhood retail outlet located in Kupari Market, Balasore. The store serves the surrounding residential community and sells a wide range of daily-use household products such as rice, pulses, oil, packaged snacks, and toiletries. Operating with a small team, the store handles around 50–70 transactions daily but relies mostly on manual observation and experience for making business decisions related to restocking, promotions, and inventory management.

Currently, the store maintains sales and transaction records but lacks a structured approach to analyze them for insights. As a result, co-purchase patterns, buying trends, and inventory inefficiencies remain undiscovered. This leads to frequent stock imbalances and missed opportunities for product bundling and cross-selling. The absence of data-driven analysis highlights the need for a practical business data management approach that can convert sales information into valuable business intelligence.

To address this, the project applies Market Basket Analysis (MBA) using the Apriori algorithm to uncover product associations from three months of transactional data. The analysis will identify frequently co-purchased products, guiding recommendations on shelf placement, promotional bundles, and inventory planning. Implemented through Python and Excel, this project will demonstrate how structured data management can empower small retailers to make evidence-based decisions, improve operational efficiency, and enhance overall profitability.

2. Organisation Background



Nana Grocery Store is a small neighborhood retail outlet established in 2015 by Mr. Balaram Taidi. Located in Kupari Market, Balasore, the store primarily serves the surrounding residential community. It offers a broad range of daily-use household products such as rice, pulses, oil, packaged snacks, cleaning supplies, and toiletries. The store operates with a small team of three employees, including the owner, and handles an average of 50–70 transactions per day.

The business follows a conventional retail model with manual inventory management and limited digital recordkeeping. While QR-based payment systems are used, the data from these transactions is not analyzed for business insights. Most operational decisions—such as restocking, purchasing, and promotions—are based on the owner's experience and direct observation.

Nana Grocery Store aims to provide convenience, fair pricing, and personalized service to its customers. Its consistent sales activity and manageable scale make it an ideal environment for implementing data management techniques that can enhance decision-making and improve business efficiency.

3. Problem Statement

3.1 Lack of a structured system to analyze customer purchasing behavior. Although Nana Grocery Store records daily transactions, the data is not utilized to identify relationships between products frequently bought together. This prevents the retailer from recognizing co-purchase patterns, leading to missed opportunities for product bundling, inefficient shelf placement, and limited sales growth.

3.2 Absence of effective data management and analytical utilization. The store's sales records remain unorganized and underused, restricting insights into customer buying trends and accurate demand forecasting. Without a systematic data management framework, decisions are based on intuition rather than evidence, resulting in inefficient inventory planning, poorly targeted promotions, and reduced profitability.

4. Background of the Problem

Nana Grocery Store records its daily sales and transactions but lacks an organized system for analyzing this data to improve decision-making. Each sale contains valuable details about customer preferences, buying habits, and product combinations, yet this information remains unused. Decisions related to inventory restocking, shelf arrangement, and promotions are made primarily through observation and experience. This unstructured approach often results in operational inefficiencies such as overstocking less-demanded products, running out of high-demand items, and missing opportunities for effective marketing or product bundling.

Although the store uses digital payment systems and maintains electronic records, the data is scattered across multiple sources without proper categorization or integration. The absence of structured data management prevents the identification of frequent product combinations and purchasing trends over time. As a result, management decisions rely on intuition rather than insights derived from evidence, restricting the store's potential for growth and innovation.

The significance of addressing this problem lies in improving both operational and financial efficiency. By implementing systematic data organization and applying analytical techniques like Market Basket Analysis, the store can convert raw transactional data into actionable insights. This would enable smarter inventory planning, better shelf design, and data-driven marketing strategies. Establishing a structured data management framework would ultimately help Nana Grocery Store transition from reactive management to proactive, evidence-based decision-making, leading to improved profitability and customer satisfaction.

5. Problem Solving Approach

5.1 Methods:

The project will be carried out using a structured analytical workflow that combines Exploratory Data Analysis (EDA) and Market Basket Analysis (MBA). The workflow begins with EDA, where I will use Excel and Python to summarize and visualize the sales data. This will help me understand product-level performance, sales frequency, and seasonal variations. These insights will guide how I select parameters for the Apriori algorithm, such as minimum support and confidence thresholds.

Next, I will perform Market Basket Analysis using the Apriori algorithm to identify products that are frequently purchased together. The algorithm will be applied iteratively, refining rule thresholds until meaningful and actionable patterns emerge. These association rules will directly inform decisions about shelf placement, combo offers, and cross-promotion opportunities within the store.

Finally, I will apply supplementary descriptive techniques, such as ABC classification, to categorize products by revenue contribution and prioritize recommendations that are both analytically strong and business-relevant.

5.2 Data Collection:

I will collect three months of transactional sales data from Nana Grocery Store, consisting of approximately 4,500–6,000 transactions. Each transaction will include details such as Date, Bill Number, Product Name, Quantity, and Price.

Date will be used to observe time-based patterns.

Bill Number will define a unique transaction, grouping items purchased together.

Product Name and Quantity will allow analysis of which items co-occur in baskets.

Price will enable identification of high-value combinations.

The dataset will be compiled from existing sales registers and digital payment records. I will manually clean and organize the data in Excel, ensuring product names and entries are consistent. Three months of data provides a sufficient sample to identify recurring purchase patterns without seasonal bias.

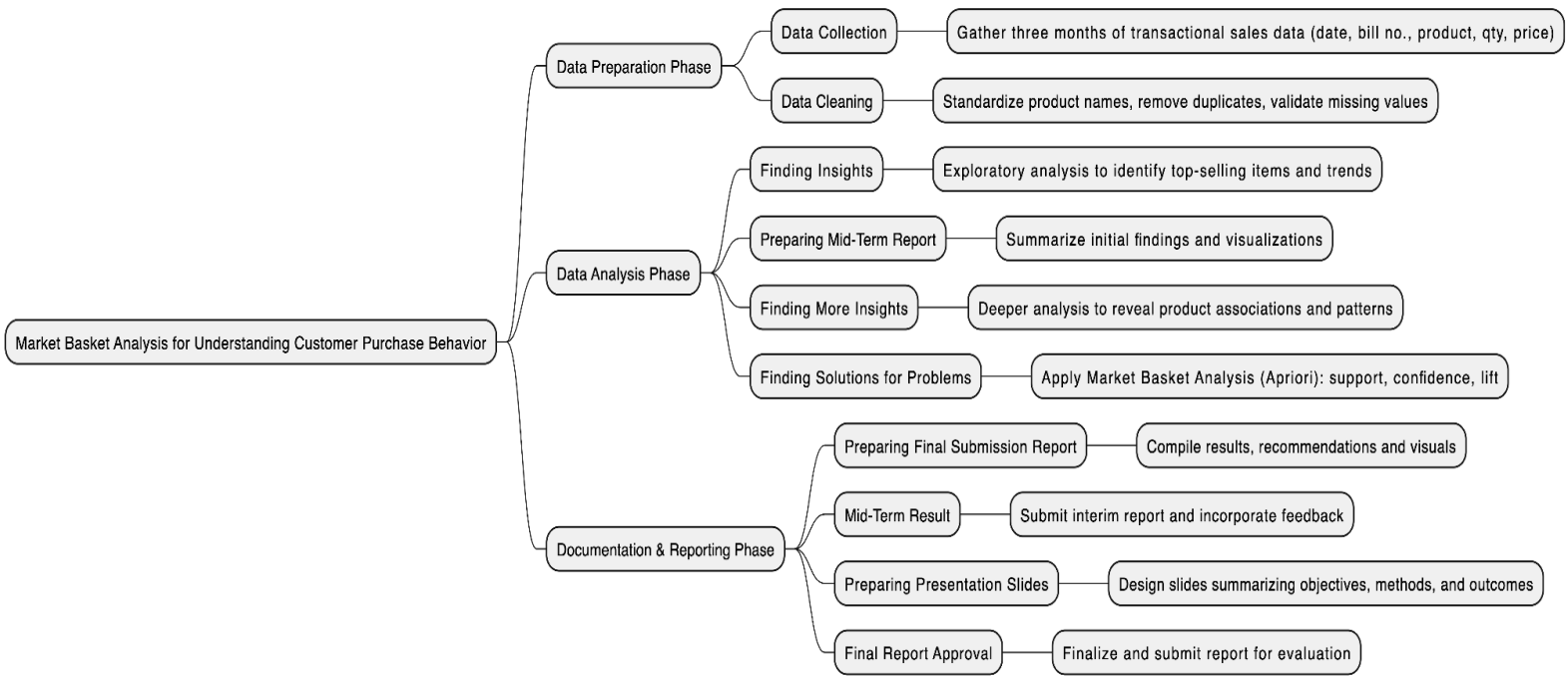
5.3 Analysis Tools:

For analysis, I will use Python (libraries: pandas and mlxtend) to run the Apriori algorithm and extract association rules. Python allows automation of repetitive computations and accurate calculation of support, confidence, and lift metrics. I will then use Excel to summarize these results in a readable format with pivot tables and simple dashboards. Visualizations—such as bar charts, heatmaps, and network graphs—will be created to clearly show product associations.

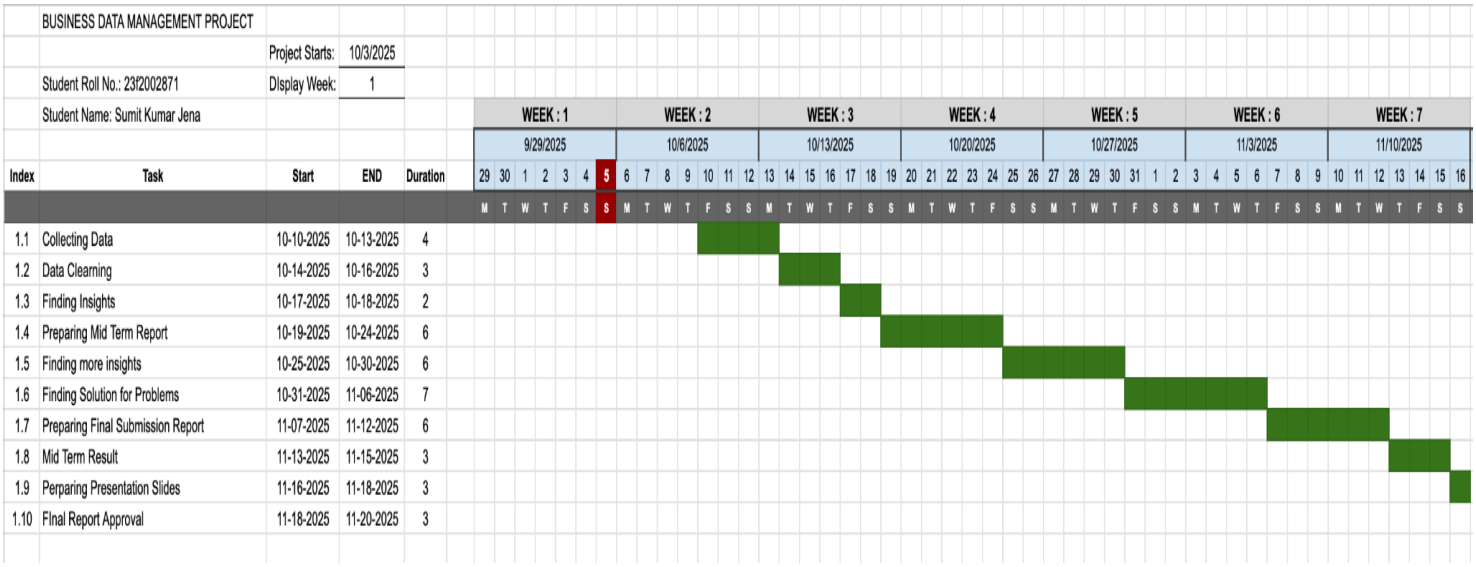
The insights will be compiled into an implementation report for the store owner, including shelf layout suggestions, combo product ideas, and data-backed inventory adjustments. This ensures that analytical findings are directly translated into practical business improvements.

6.Expected Timeline

6.1 Work Breakdown Structure



6.2 Gantt Chart



	BUSINESS DATA MANAGEMENT PROJECT			
		Project Starts:	10/3/2025	
	Student Roll No.: 23f2002871	Display Week:	1	
	Student Name: Sumit Kumar Jena			

					WEEK : 1							WEEK : 2							WEEK : 3							WEEK : 4							WEEK : 5							WEEK : 6							WEEK : 7						
					9/29/2025							10/6/2025							10/13/2025							10/20/2025							10/27/2025							11/3/2025							11/10/2025						
Index	Task	Start	END	Duration	29	30	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
					M	T	W	T	F	S	S	M	T	W	T	F	S	S	M	T	W	T	F	S	S	M	T	W	T	F	S	S	M	T	W	T	F	S	S	M	T	W	T	F	S	S	M	T	W	T	F	S	S
1.1	Collecting Data	10-10-2025	10-13-2025	4																																																	
1.2	Data Cleaning	10-14-2025	10-16-2025	3																																																
1.3	Finding Insights	10-17-2025	10-18-2025	2																																																
1.4	Preparing Mid Term Report	10-19-2025	10-24-2025	6																																															
1.5	Finding more insights	10-25-2025	10-30-2025	6																																															
1.6	Finding Solution for Problems	10-31-2025	11-06-2025	7																																															
1.7	Preparing Final Submission Report	11-07-2025	11-12-2025	6																																															
1.8	Mid Term Result	11-13-2025	11-15-2025	3																																															
1.9	Perparing Presentation Slides	11-16-2025	11-18-2025	3																																															
1.10	FInal Report Approval	11-18-2025	11-20-2025	3																																															

7. Expected Outcome

The project is expected to deliver a structured and data-driven understanding of customer purchase behavior at Nana Grocery Store. By analyzing three months of transactional data using Market Basket Analysis, the project will generate practical insights that directly contribute to improving operational efficiency and sales performance.

The primary deliverables will include:

- A cleaned and well-structured dataset suitable for future use.
- Association-rule tables identifying frequently co-purchased products.
- Visual reports and charts illustrating product relationships and sales trends.
- A concise implementation report recommending shelf arrangements, combo offers, and inventory adjustments.

From these deliverables, several key insights are anticipated. The analysis will highlight strong product associations that support cross-selling and bundling strategies. It will also reveal demand patterns that enable more accurate stock forecasting. These insights, derived entirely from data rather than intuition, will empower the store owner to make informed, evidence-based decisions. Ultimately, the project will demonstrate how small retailers can apply Business Data Management techniques to convert raw sales data into actionable strategies that enhance customer satisfaction and profitability.